

Planning Summary – Powerful Teaching by Pooja Agawal, Patrice Bain

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Content	Do This/Remember This
Introduction: You will gain: deep understanding of powerful teaching strategies , knowledge of key findings of cognitive science , insight into effective implementation , and tools to share with students. All with minimal additional prep/grading	Science of learning doesn't mean reinventing the wheel ... or take a lot of time
Chapter 1 Discover the Power Behind Power Tools (PTs): Cognitive scientists use 'Encoding/ Storage/ Retrieval' as a simplified model of learning, PTs are Retrieval Practice, Spacing, Interleaving, Feedback-Driven Metacognition , all backed by findings from the science of leaning . Distinguish short term vs long term learning (LTM) for you and the students. Resources at https://www.retrievalpractice.org/powerful-teaching/	In learning easier isn't better Teach students that struggle is helpful/desirable difficulties
Chapter 2 Build a Foundation with Retrieval Practice: Encoding (E) = initial learning , Storage (S) = how long retained , Retrieval (R) = bringing info to mind . Easy to focus on E but R practice (RP) is more effective than many strategies (re-reading/highlighting - students (Ss) often overestimate their power), backed by research in classrooms, Lemov definition "RP occurs when learners recall and apply multiple examples of previously learned knowledge or skills after a period of forgetting". Can facilitate Higher Order Thinking (HOT) , RP can include HOT. RP boosts transfer to other settings (near and far). RP different from formative/summative assessment (though similarities)	Consider E/S/R when planning Counsell students on strategies Use definition when planning – RP needs delay so there is effort Make some RP HOT Make it low or no stakes
Chapter 3 Empower Teaching with Retrieval Practice Strategies: Strategies are: Brain dumps – everything they know + think/pair/share, Two Things – things you've learned today/before or examples in real life etc., Retrieve Taking – take notes after the presentation/ review/ amend , Retrieval Guides – booklets used in the same way, Mini/Weekly Quizzes – with variety of questions. Systems to use are: Bell work/exit tickets, Colored Index Cards (like clickers), Mini-whiteboards . Make sure ALL Ss engaged. Research – boosts learning with all types of question , effect of pre-testing on LTM unclear, RP better than concept mapping	Use a variety of RP strategies Keep it short Build in routines Use a strategy that engages all Ss Mix it up Start small, try things out
Chapter 4 Energize Learning with Spacing and Interleaving: Cramming works in the short-term only. Spacing (gaps between retrieval) strategies: Pre-test, Blast from the Past (Turn & Talk about a previous topic), Big Basket Quiz (BBQ – lots of questions from all parts of the course, choose 10). Timing: best practice = a few days after teaching, then weeks, then months. Spacing = saving time – they remember more, quicker. Interleaving: = mixing up with spacing . Promotes discrimination, increases learning in the middle of the unit . Strategies: Use dice/pulling out questions from hat/fast paced quiz on multiple topics	Plan for spacing Spiraling is not spacing Forgetting (just), is key to remembering Set homework the week <i>after</i> they've learned it Care with interleaving – plan it
Chapter 5 Engage Students with Feedback-Driven Metacognition: ...so students are aware of what they know and don't know . Illusions –students can predict what they got correct/confidence in sync with learning. Feedback: right/wrong answers, elaborative (good by time-consuming, immediately OR delayed, normalize making mistakes in low stakes testing. 4 steps of metacognition: identify what you do/don't know, do what you know, look up what you don't, check. Encourage reflection in retrieval.	Give feedback so confidence in sync with learning Combine spacing by adding delay to feedback Examine confidence with 'stand on a line'

Chapter 6 Combine Power Tools and Harness Your Toolbox: Reviewing is <i>not</i> retrieving, cumulative exams are <i>not</i> spacing + interleaving. Teacher toolbox: essential question, need to know, retrieving, spacing, interleaving, metacognition. Student toolbox: mnemonics. Combine PTs: brain dumps+last lesson+turn/talk, write+leave+retrieve. Tech doesn't necessarily improve learning – choose wisely (clickers can increase gamification not learning)	Your course design drives student toolboxes Use 'Power Tickets' Combine strategies Make conscious choice of tech
Chapter 7 Keeping It Real: Use Power Tools to Tackle Challenges, Not Add to Them: Worrisome things: prep time, grading time, coverage, grades, inclusion, cost, complexity, overlap, support. Answers: no more time prep/grading (sometimes less), retrieval = faster as teaching = learning, initial student struggle = better learning, PT really help diverse learners + boost self esteem, no cost, start small+ expand, beware neuroscience	Lots of the PT reduce time for prep/grading Start with a few tools Check neuromyths
Chapter 8 Foster a Supportive Environment: Use Power Tools to Reduce Anxiety and Strengthen Community: Sources of student anxiety: infrequent testing, high stakes, correct or incorrect. Create support by emphasizing: it takes time + mistakes + risks + help to learn. Build trust by: modelling – make mistakes, explaining how we learn, redefining 'struggle' as good, change 'smart' to 'mastery'. Reduce anxiety by: regular retrieval, multiple opportunities to add to grade, eliminate competition, no right/wrong questions	Reduce high stakes testing to minimum Emphasize importance of making mistakes PT reduce anxiety Avoid games
Chapter 9 Spark Conversations with Students About the Science of Learning: Empower students by: sparking conversation/modelling PTs/explain how PTs work/encourage use inside AND outside classroom/encourage planning+implementing+reflecting on their PTs. We are teaching them how to learn, learning is a joint responsibility: teacher + encoding, student + retrieval. Four key phrases: let's have a pointless conversation/let's learn OUTformation not just Information/what did you learn yesterday/let's flip retrieval from – to +. Share why it works: use it or lose it/desirable difficulties/know vs don't know. Ownership by: having names for the PTs, explaining why some things <i>don't</i> work, review all the study strategies with reasons, conversations about planning+reflection	Talk to students about PTs Don't impose, encourage Remember all this is new – needs explaining Try the 4 key phrases Consciously model the PTs Make up names for the PTs Have conversations about in/out of class
Chapter 10 Spark Conversations with Parents About the Science of Learning: Opportunities: PT conferences/seminars/email/informal. What to say: mini-lesson on PTs, strategies: tell me three (things you learned), flashcard use. Research snapshots to share: retrieval practice, spacing, interleaving, metacognition, combining PTs, each with research evidence	Introduce parents to PTs formally or informally Give concrete strategies they can use Explain science behind it all
Chapter 11 Powerful Professional Development for Teachers and Leaders: No quick fixes in policy making BUT policy makers/future teachers/principles + superintendents must use evidence from cognitive science. Goals: an informed teacher is a powerful leader/lead your own PD/lead PD for others using the PTs/reflect to spark powerful conversations/use PT implementation checklists	Spread the word! When you know better, do better! Eradicate 'one and done' Use book clubs
Chapter 12 Do It Yourself Retrieval Guide: Get the book and fill in this section to check your understanding	Use PTs to assess your understanding of PTs!
Conclusion: Re image on cover: four sparks are the PTs, the fifth spark is YOU!	Unleash your PTs!